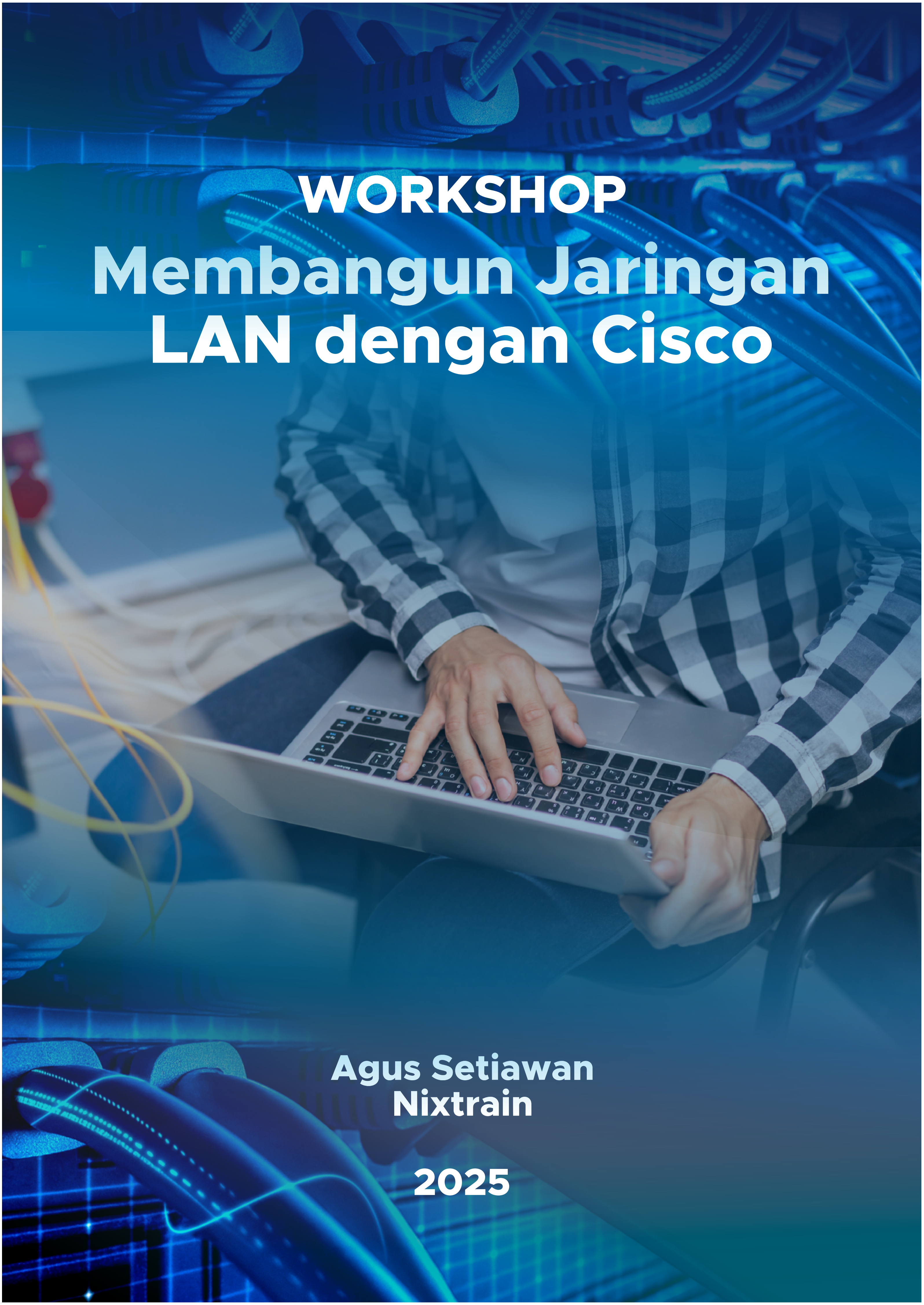


WORKSHOP
Membangun Jaringan LAN dengan Cisco

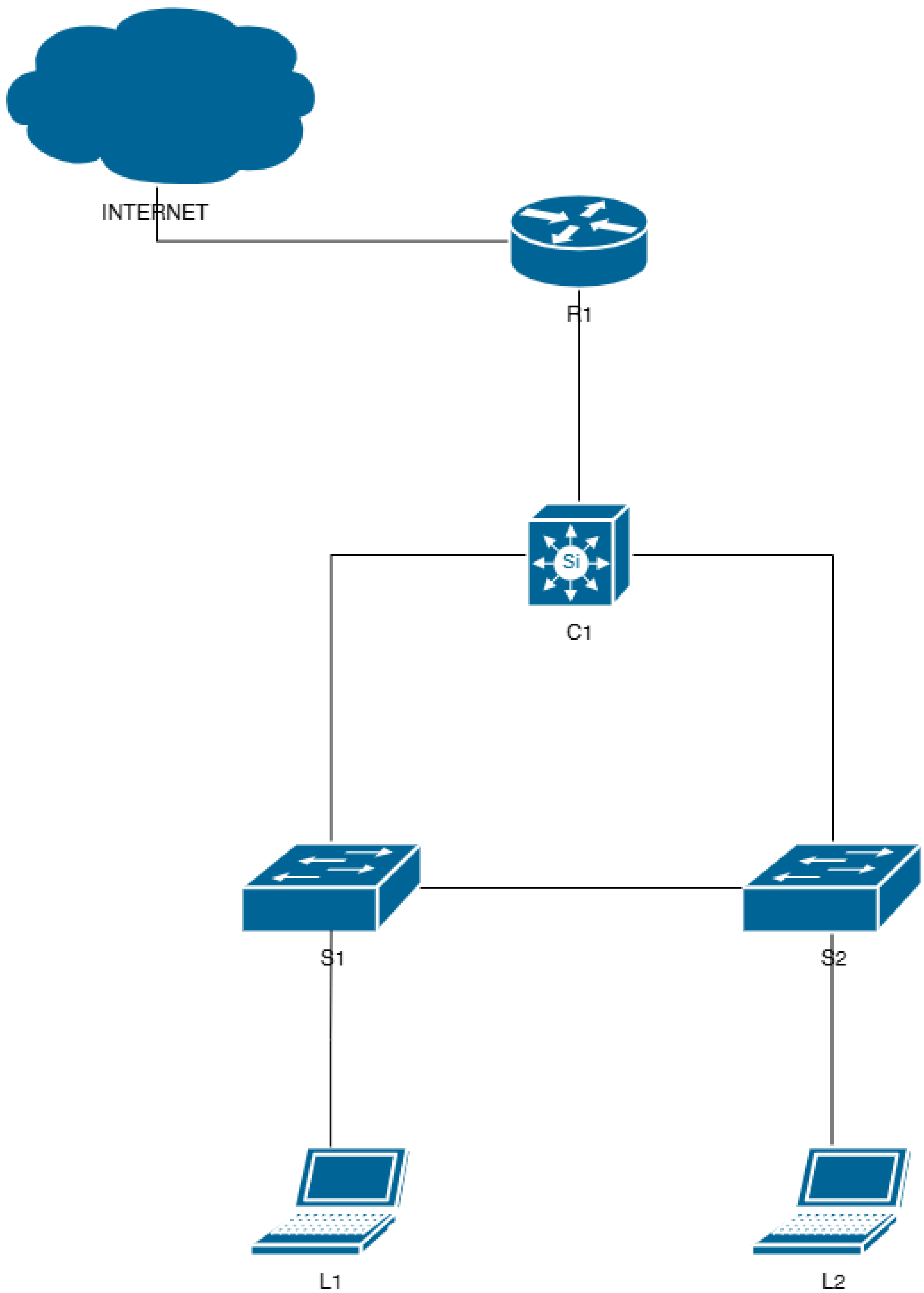
Agus Setiawan
Nixtrain

2025

Daftar Isi

Lab 1. Topologi Jaringan LAN, Device Information dan IP Addressing

Topology



Device Information

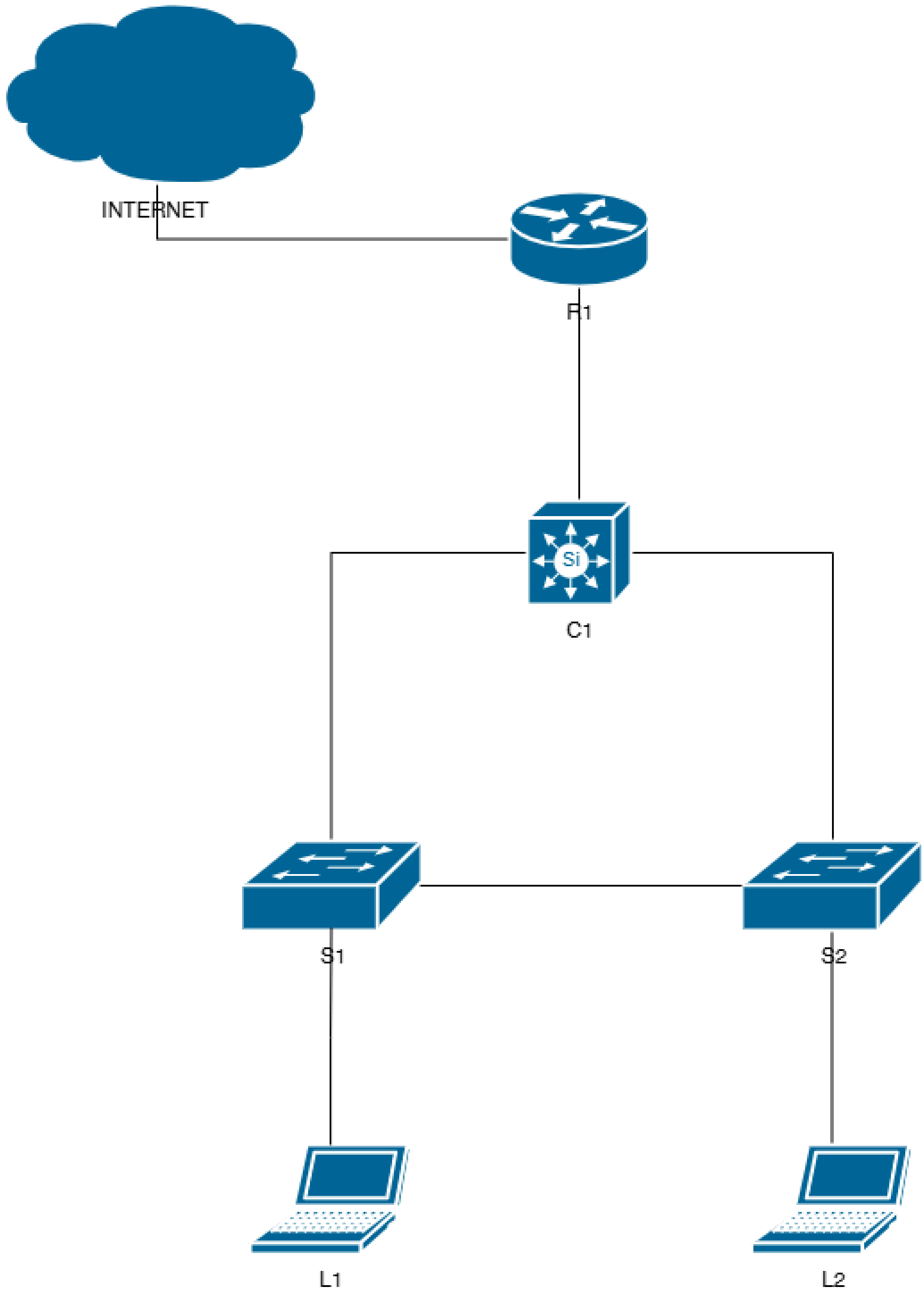
Hostname	Device Model	Software Version
Internet	N/A	N/A
R1	Cisco 1841	C1841-ADVENTERPRISEK9-M Version 12.4(9)T1
C1	Cisco WS-C3750V2-48PS	C3750-IPBASEK9-M Version 12.2(50)SE5
S1	Cisco WS-C2950-24	C2950-I6K2L2Q4-M Version 12.1(22)EA13
S2	Cisco WS-C2950-24	C2950-I6Q4L2-M Version 12.1(22)EA4
L1	Laptop	Windows 11
L2	Laptop	Windows 11

IP Addressing

Segmen	IP Addressing
Internet	DHCP
R1-C1	10.1.2.0/24
Loopback R1	1.1.1.1/32
Loopback C1	2.2.2.2/32
VLAN 10	192.168.10.0/24
VLAN 20	192.168.20.0/24

Lab 2. Basic Configuration Switch dan Router

Topology



Konfigurasi

- Konfigurasi basic switch dan router sesuai standard lab

R1

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#enable secret ciscosec
R1(config)#
R1(config)#banner motd $
Enter TEXT message.  End with the character '$'.
Authorized users only
$
R1(config)#
R1(config)#service password-encryption
R1(config)#
R1(config)#line console 0
R1(config-line)#password ciscocon
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#
R1(config-line)#line vty 0 4
R1(config-line)#password ciscovty
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#
R1(config-line)#do write
Building configuration...
[OK]
R1(config-line)#
```

C1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname C1
C1(config)#
C1(config)#enable secret ciscosec
C1(config)#
C1(config)#service password-encryption
C1(config)#
C1(config)#banner motd $
```

```
Enter TEXT message.  End with the character '$'
Authorized users only
$
C1(config)#
C1(config)#line console 0
C1(config-line)#password ciscocon
C1(config-line)#login
C1(config-line)#logging synchronous
C1(config-line)#
C1(config-line)#line vty 0 4
C1(config-line)#password ciscovty
C1(config-line)#login
C1(config-line)#logging synchronous
C1(config-line)#
C1(config-line)#do write
Building configuration...
[OK]
C1(config-line)#
```

S1

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#
S1(config)#enable secret ciscosec
S1(config)#
S1(config)#service password-encryption
S1(config)#
S1(config)#banner motd $
Enter TEXT message.  End with the character '$'.
Authorized users only
$
S1(config)#
S1(config)#line console 0
S1(config-line)#password ciscocon
S1(config-line)#login
S1(config-line)#logging synchronous
S1(config-line)#
S1(config-line)#line vty 0 4
S1(config-line)#password ciscovty
S1(config-line)#login
S1(config-line)#logging synchronous
S1(config-line)#
S1(config-line)#do write
Building configuration...
[OK]
S1(config-line)#
```

S2

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname S2
S2(config)#
S2(config)#enable secret ciscosec
S2(config)#
S2(config)#service password-encryption
S2(config)#
S2(config)#banner motd $
Enter TEXT message.  End with the character '$'.
Authorized users only
$
S2(config)#
S2(config)#line console 0
S2(config-line)#password ciscocon
S2(config-line)#login
S2(config-line)#logging synchronous
S2(config-line)#
S2(config-line)#line vty 0 4
S2(config-line)#password ciscovty
S2(config-line)#login
S2(config-line)#logging synchronous
S2(config-line)#
S2(config-line)#do write
Building configuration...
[OK]
S2(config-line)#
```

- Inputkan IP Addressing sesuai table addressing

R1

```
R1(config)#int lo0
R1(config-if)#ip add 1.1.1.1 255.255.255.255
R1(config-if)#
R1(config-if)#int f0/0
R1(config-if)#description Koneksi-Internet
R1(config-if)#ip address dhcp
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#int f0/1
R1(config-if)#description Koneksi-C1
R1(config-if)#ip add 10.1.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
```

C1

```
C1(config)#int f1/0/1
C1(config-if)#no switchport
C1(config-if)#description Koneksi-R1
C1(config-if)#ip add 10.1.2.2 255.255.255.0
C1(config-if)#no shutdown
C1(config-if)#
C1(config-if)#int lo0
C1(config-if)#ip add 2.2.2.2 255.255.255.255
C1(config-if)#
C1(config-if)#ip routing
C1(config)#
```

R1 dan C1 sudah di konfigurasi IP address pada masing-masing interface fisik maupun loopbacknya. Pada lab ini, S1 dan S2 tidak disetting IP address, sehingga untuk manajemen perangkat harus melalui kabel console langsung.

Verifikasi

Untuk verifikasi basic konfigurasi dapat menggunakan command berikut ini :

- `show running-config`
- `show ip interface brief`
- `show interface`
- `show interface description`
- ping end-to-end connection device

R1

```
R1#
R1#show run
Building configuration...

Current configuration : 876 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$HhVJ$wzkVas3eMT5/ZL5/wGbQo0
```

```
!
no aaa new-model
!
resource policy
!
ip cef
!
!
!
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 description Koneksi-Internet
 ip address dhcp
 duplex auto
 speed auto
!
interface FastEthernet0/1
 description Koneksi-C1
 ip address 10.1.2.1 255.255.255.0
 duplex auto
 speed auto
 mpls ip
!
!
!
ip http server
no ip http secure-server
!
!
!
!
!
!
!
control-plane
!
!
banner motd ^C
Authorized users only
^C
```

```
!
line con 0
 password 7 121A0C0411040F0B24
 logging synchronous
 login
line aux 0
line vty 0 4
 password 7 110A1016141D1D181D
 logging synchronous
 login
!
scheduler allocate 20000 1000
end

R1#
R1#show ip int brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            192.168.1.18    YES DHCP   up
up
FastEthernet0/1            10.1.2.1        YES manual up
up
Loopback0                  1.1.1.1         YES manual up
up
R1#
R1#show interfaces fa0/1
FastEthernet0/1 is up, line protocol is up
  Hardware is Gt96k FE, address is d057.4c73.b4b1 (bia
d057.4c73.b4b1)
  Description: Koneksi-C1
  Internet address is 10.1.2.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, 100BaseTX/FX
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:18, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total
output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     2582 packets input, 624920 bytes
     Received 2578 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog
     0 input packets with dribble condition detected
     10479 packets output, 886146 bytes, 0 underruns
     0 output errors, 0 collisions, 3 interface resets
```

```
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
R1#
R1#show int desc
Interface                      Status         Protocol
Description
Fa0/0                          up             up
Koneksi-Internet
Fa0/1                          up             up
Koneksi-C1
Lo0                            up             up
R1#
```

C1

```
C1#
C1#show run
Building configuration...

Current configuration : 2504 bytes
!
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname C1
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$VxNq$ZsS3Pox7576BgrUgekMiO1
!
no aaa new-model
switch 1 provision ws-c3750v2-48ps
system mtu routing 1500
ip subnet-zero
ip routing
!
!
!
!
!
!
!
!
!
```

```
!
spanning-tree mode pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
!
!
interface Loopback0
 ip address 2.2.2.2 255.255.255.255
!
interface FastEthernet1/0/1
 no switchport
 ip address 10.1.2.2 255.255.255.0
!
interface FastEthernet1/0/2
!
interface FastEthernet1/0/3
!
interface FastEthernet1/0/4
!
interface FastEthernet1/0/5
!
interface FastEthernet1/0/6
!
interface FastEthernet1/0/7
!
interface FastEthernet1/0/8
!
interface FastEthernet1/0/9
!
interface FastEthernet1/0/10
!
interface FastEthernet1/0/11
!
interface FastEthernet1/0/12
!
interface FastEthernet1/0/13
!
interface FastEthernet1/0/14
!
interface FastEthernet1/0/15
!
interface FastEthernet1/0/16
!
interface FastEthernet1/0/17
!
interface FastEthernet1/0/18
!
interface FastEthernet1/0/19
!
interface FastEthernet1/0/20
```

```
!
interface FastEthernet1/0/21
!
interface FastEthernet1/0/22
!
interface FastEthernet1/0/23
!
interface FastEthernet1/0/24
!
interface FastEthernet1/0/25
!
interface FastEthernet1/0/26
!
interface FastEthernet1/0/27
!
interface FastEthernet1/0/28
!
interface FastEthernet1/0/29
!
interface FastEthernet1/0/30
!
interface FastEthernet1/0/31
!
interface FastEthernet1/0/32
!
interface FastEthernet1/0/33
!
interface FastEthernet1/0/34
!
interface FastEthernet1/0/35
!
interface FastEthernet1/0/36
!
interface FastEthernet1/0/37
!
interface FastEthernet1/0/38
!
interface FastEthernet1/0/39
!
interface FastEthernet1/0/40
!
interface FastEthernet1/0/41
!
interface FastEthernet1/0/42
!
interface FastEthernet1/0/43
!
interface FastEthernet1/0/44
!
interface FastEthernet1/0/45
!
interface FastEthernet1/0/46
```

```
!
interface FastEthernet1/0/47
!
interface FastEthernet1/0/48
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip http server
ip http secure-server
!
!
control-plane
!
banner motd ^C
Authorized users only
^C
!
line con 0
 password 7 1511021F0725282B26
 logging synchronous
 login
line vty 0 4
 password 7 13061E010803123E32
 logging synchronous
 login
line vty 5 15
 login
!
end

C1#
C1#show ip int brief
Interface              IP-Address      OK? Method Status
Protocol
Vlan1                  unassigned      YES unset
administratively down down
FastEthernet1/0/1      10.1.2.2        YES manual up
up
FastEthernet1/0/2      unassigned      YES unset  down
down
FastEthernet1/0/3      unassigned      YES unset  down
```

```
down
FastEthernet1/0/4      unassigned      YES unset  down
down
FastEthernet1/0/5      unassigned      YES unset  down
down
FastEthernet1/0/6      unassigned      YES unset  down
down
FastEthernet1/0/7      unassigned      YES unset  down
down
FastEthernet1/0/8      unassigned      YES unset  down
down
FastEthernet1/0/9      unassigned      YES unset  down
down
FastEthernet1/0/10     unassigned      YES unset  down
down
FastEthernet1/0/11     unassigned      YES unset  down
down
FastEthernet1/0/12     unassigned      YES unset  down
down
FastEthernet1/0/13     unassigned      YES unset  down
down
FastEthernet1/0/14     unassigned      YES unset  down
down
FastEthernet1/0/15     unassigned      YES unset  down
down
FastEthernet1/0/16     unassigned      YES unset  down
down
FastEthernet1/0/17     unassigned      YES unset  down
down
FastEthernet1/0/18     unassigned      YES unset  down
down
FastEthernet1/0/19     unassigned      YES unset  down
down
FastEthernet1/0/20     unassigned      YES unset  down
down
FastEthernet1/0/21     unassigned      YES unset  down
down
FastEthernet1/0/22     unassigned      YES unset  down
down
FastEthernet1/0/23     unassigned      YES unset  down
down
FastEthernet1/0/24     unassigned      YES unset  down
down
FastEthernet1/0/25     unassigned      YES unset  down
down
FastEthernet1/0/26     unassigned      YES unset  down
down
FastEthernet1/0/27     unassigned      YES unset  down
down
FastEthernet1/0/28     unassigned      YES unset  down
down
FastEthernet1/0/29     unassigned      YES unset  down
```

```
down
FastEthernet1/0/30     unassigned      YES unset  down
down
FastEthernet1/0/31     unassigned      YES unset  down
down
FastEthernet1/0/32     unassigned      YES unset  down
down
FastEthernet1/0/33     unassigned      YES unset  down
down
FastEthernet1/0/34     unassigned      YES unset  down
down
FastEthernet1/0/35     unassigned      YES unset  down
down
FastEthernet1/0/36     unassigned      YES unset  down
down
FastEthernet1/0/37     unassigned      YES unset  down
down
FastEthernet1/0/38     unassigned      YES unset  down
down
FastEthernet1/0/39     unassigned      YES unset  down
down
FastEthernet1/0/40     unassigned      YES unset  down
down
FastEthernet1/0/41     unassigned      YES unset  down
down
FastEthernet1/0/42     unassigned      YES unset  down
down
FastEthernet1/0/43     unassigned      YES unset  down
down
FastEthernet1/0/44     unassigned      YES unset  down
down
FastEthernet1/0/45     unassigned      YES unset  down
down
FastEthernet1/0/46     unassigned      YES unset  down
down
FastEthernet1/0/47     unassigned      YES unset  up
up
FastEthernet1/0/48     unassigned      YES unset  up
up
GigabitEthernet1/0/1   unassigned      YES unset  down
down
GigabitEthernet1/0/2   unassigned      YES unset  down
down
GigabitEthernet1/0/3   unassigned      YES unset  down
down
GigabitEthernet1/0/4   unassigned      YES unset  down
down
Loopback0              2.2.2.2         YES manual up
up
C1#
C1#show int
Vlan1 is administratively down, line protocol is down
```

```
  Hardware is EtherSVI, address is 10bd.18fb.a9c0 (bia
10bd.18fb.a9c0)
  MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive not supported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:02:07, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total
output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     1141 packets input, 375663 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 1 interface resets
     0 output buffer failures, 0 output buffers swapped out
FastEthernet1/0/1 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is 10bd.18fb.a9c1 (bia
10bd.18fb.a9c1)
  Internet address is 10.1.2.2/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
  input flow-control is off, output flow-control is
unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:02, output 00:00:02, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total
output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     10434 packets input, 927193 bytes, 0 no buffer
     Received 7423 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog, 7420 multicast, 0 pause input
     0 input packets with dribble condition detected
     23790 packets output, 2522937 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier, 0 PAUSE output
```

```
     0 output buffer failures, 0 output buffers swapped out
  ……………..
C1#
C1#show int desc
Interface                      Status         Protocol
Description
Vl1                            admin down     down
Fa1/0/1                        up             up
Koneksi-R1
Fa1/0/2                        down           down
Fa1/0/3                        down           down
```

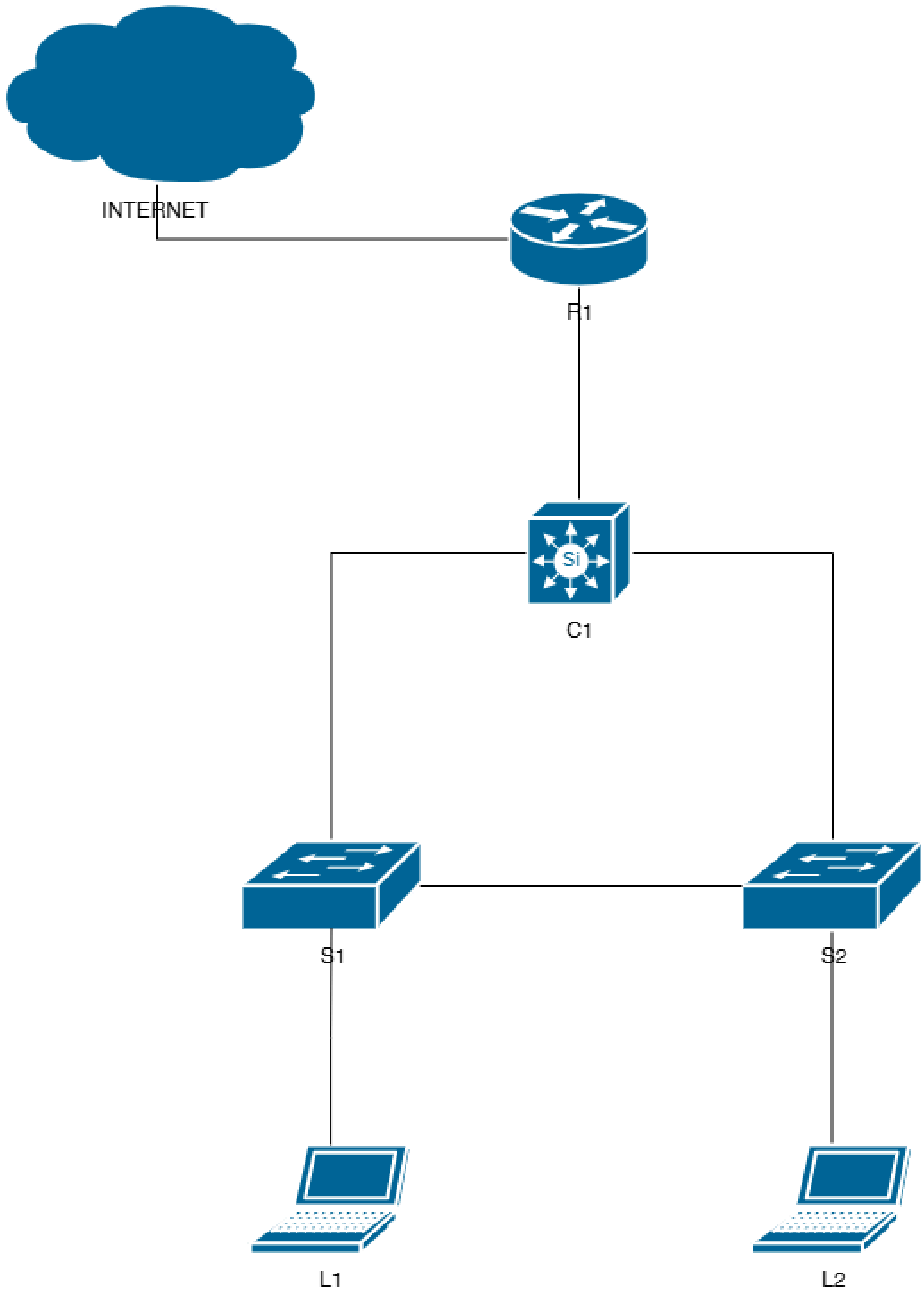
Pastikan ping antara R1 dan C1 berhasil.

C1

```
C1#
C1#ping 10.1.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.1.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/9 ms
C1#
C1#
```

Lab 3. VLAN dan Trunk

Topology



Konfigurasi

- Konfigurasi VLAN 10 (IT) dan VLAN 20 (NOC) pada C1, S1, S2

C1, S1, S2

```
X(config)#vlan 10
X(config-vlan)#name IT
X(config-vlan)#vlan 20
X(config-vlan)#name NOC
```

- Konfigurasi port membership VLAN pada S1 dan S2

S1,S2

```
X(config-if-range)#
X(config-if-range)#int range fa0/2 - 10
X(config-if-range)#switchport mode access
X(config-if-range)#switchport access vlan 10
X(config-if-range)#
X(config-if-range)#int range fa0/11 - 20
X(config-if-range)#switchport mode access
X(config-if-range)#switchport access vlan 20
X(config-if-range)#
```

- Konfigurasi trunk antar switch

C1

```
C1(config-vlan)#
C1(config-vlan)#do show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
S2               Fas 1/0/48        152              S I   WS-C2950- Fas 0/1
S1               Fas 1/0/47        121              S I   WS-C2950- Fas 0/1
R1               Fas 1/0/1         158            R S I   1841      Fas 0/1
C1(config-vlan)#
```

```
C1(config-vlan)#
C1(config-vlan)#int f1/0/47
C1(config-if)#switchport trunk encapsulation dot1q
C1(config-if)#switchport mode trunk
C1(config-if)#switchport trunk allowed vlan all
C1(config-if)#
C1(config-if)#
C1(config-if)#int f1/0/48
C1(config-if)#switchport trunk encapsulation dot1q
C1(config-if)#switchport mode trunk
C1(config-if)#switchport trunk allowed vlan all
C1(config-if)#
```

S1

```
S1(config-vlan)#
S1(config-vlan)#do show cdp nei
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
S2               Fas 0/24           127          S I      WS-C2950-2Fas 0/24
C1               Fas 0/1            145        R S I      WS-C3750V2Fas 1/0/47
S1(config-vlan)#
```

```
S1(config-vlan)#
S1(config-vlan)#int f0/1
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan all
S1(config-if)#
S1(config-if)#int f0/24
S1(config-if)#switchport mode trunk
S1(config-if)#switchport trunk allowed vlan all
S1(config-if)#
```

S2

```
S2(config-vlan)#
S2(config-vlan)#do show cdp nei
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
S1               Fas 0/24           168          S I      WS-C2950-2Fas 0/24
C1               Fas 0/1            158        R S I      WS-C3750V2Fas 1/0/48
S2(config-vlan)#
```

```
S2(config-vlan)#
S2(config-vlan)#int f0/1
S2(config-if)#switchport mode trunk
S2(config-if)#switchpor trunk allowed vlan all
S2(config-if)#
S2(config-if)#int f0/24
S2(config-if)#switchport mode trunk
S2(config-if)#switchpor trunk allowed vlan all
S2(config-if)#
```

Verifikasi

Untuk verifikasi VLAN dan trunk pada switch, dapat menggunakan command berikut ini :

- `show vlan brief`
- `show int trunk`
- `show flash`
- `show run`

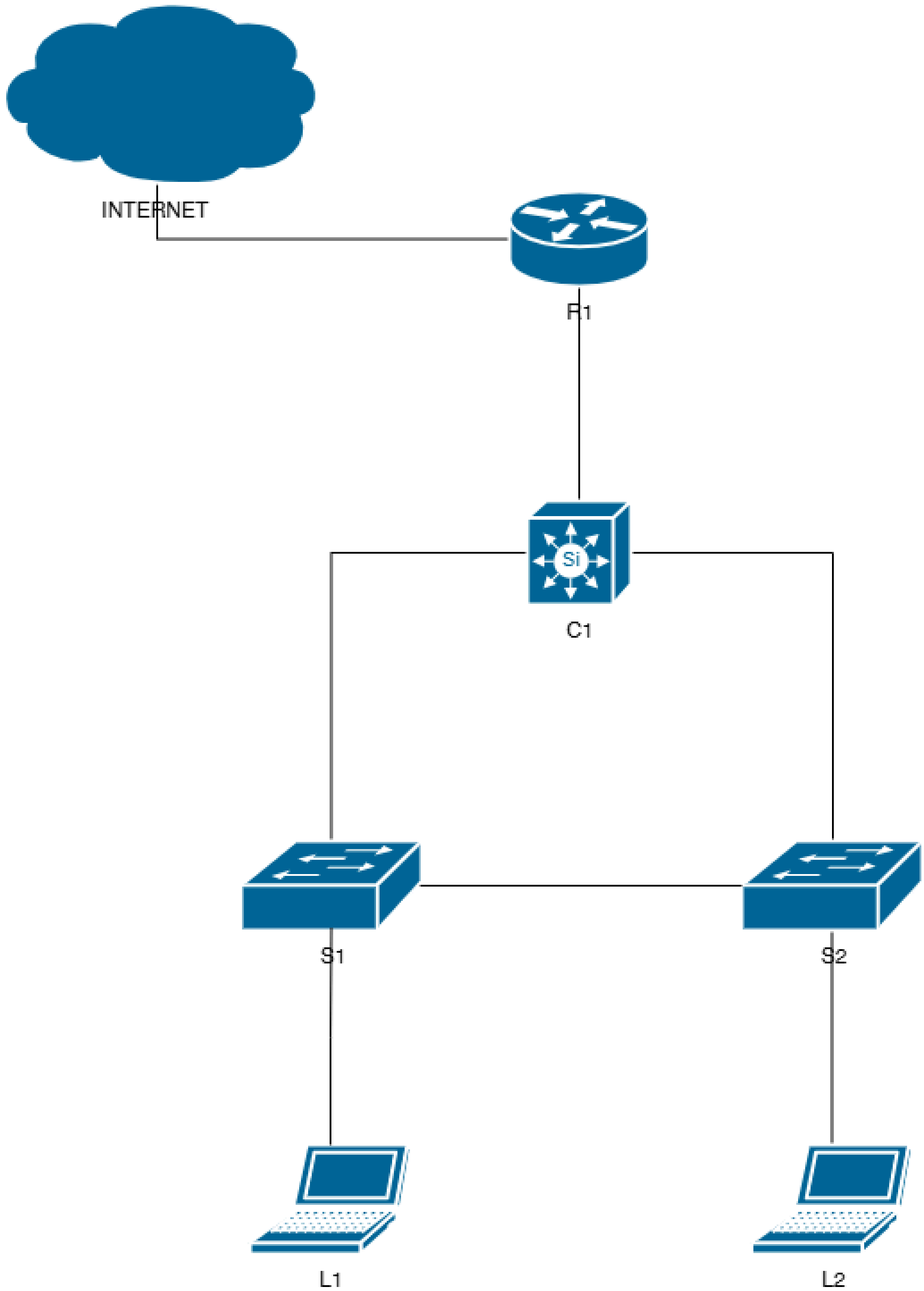
S1

```
S1#
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23
10   IT                               active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10
20   NOC                              active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
S1#
S1#show int trunk

Port        Mode          Encapsulation  Status        Native vlan
Fa0/1       on            802.1q         trunking      1
Fa0/24      on            802.1q         trunking      1

Port      Vlans allowed on trunk
Fa0/1       1-4094
Fa0/24      1-4094

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20
Fa0/24      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20
Fa0/24      1,10,20
S1#
```

Pada saat menjalankan command `show flash`, maka akan muncul file `vlan.dat` yaitu database vlan.

Lab 4. InterVLAN Routing

Topology



Konfigurasi

- Konfigurasi InterVLAN Routing di C1

Untuk membuat intervlan routing pada C1, kita menggunakan SVI (**Switch Virtual Interface**) yang berfungsi sebagai interface L3 pada masing-masing vlan.

C1
C1(config)# C1(config)#int vlan 10 C1(config-if)#ip add 192.168.10.1 255.255.255.0 C1(config-if)# C1(config-if)#int vlan 20 C1(config-if)#ip add 192.168.20.1 255.255.255.0 C1(config-if)#

Verifikasi

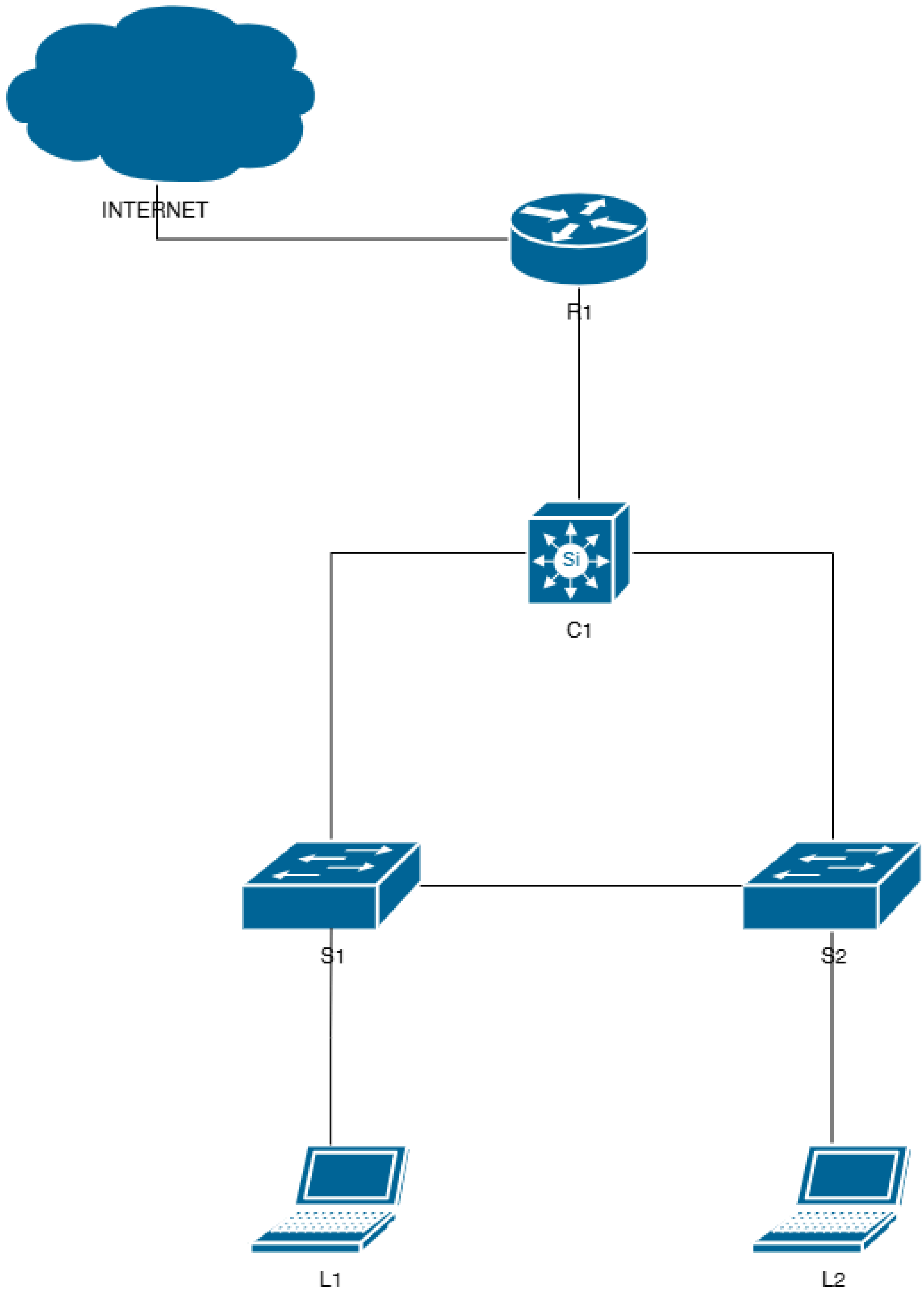
- Tampilkan routing table dan pastikan sudah muncul routing table intervlan routingnya
- Self-ping interface SVI

C1

```
C1#
C1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

     2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2 is directly connected, Loopback0
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
     10.0.0.0/24 is subnetted, 1 subnets
C       10.1.2.0 is directly connected, FastEthernet1/0/1
C1#
C1#ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/9 ms
C1#
C1#ping 192.168.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/8 ms
C1#
```

Lab 5. DHCP VLAN

Topology



Konfigurasi

- Konfigurasi DHCP VLAN pada C1

Fungsi DHCP VLAN yaitu untuk memberikan IP address secara dinamis kepada user VLAN. Pada saat user terkoneksi ke switch dengan VLAN 10, maka user akan mendapatkan IP dari DHCP Server VLAN 10.

C1
C1(config)# C1(config)#ip dhcp pool VLAN10 C1(dhcp-config)#network 192.168.10.0 255.255.255.0 C1(dhcp-config)#domain-name nixtrain.id C1(dhcp-config)#dns-server 118.98.115.77 118.98.115.70 C1(dhcp-config)#default-router 192.168.10.1 C1(dhcp-config)#lease 30 C1(dhcp-config)# C1(dhcp-config)#ip dhcp pool VLAN20 C1(dhcp-config)#network 192.168.20.0 255.255.255.0 C1(dhcp-config)#domain-name nixtrain.id C1(dhcp-config)#dns-server 118.98.115.77 118.98.115.70 C1(dhcp-config)#default-router 192.168.20.1 C1(dhcp-config)#lease 30 C1(dhcp-config)#

Verfikasi

Untuk mengetahui status DHCP dapat menggunakan command berikut ini :

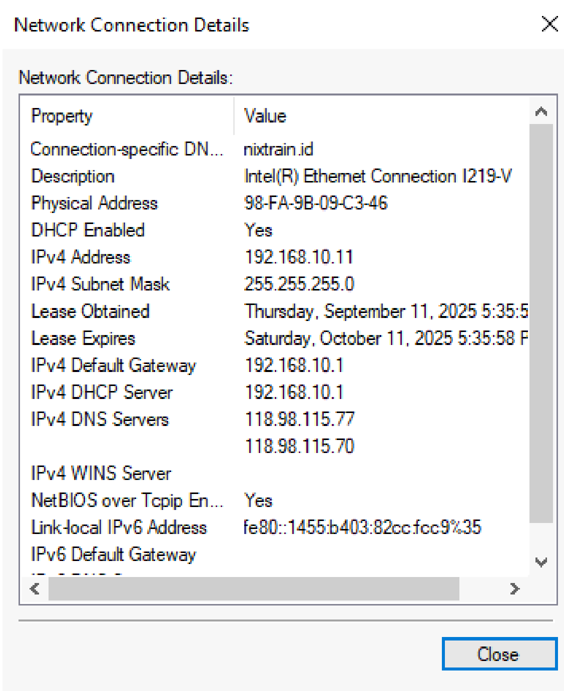
- `show ip dhcp binding`
- `show ip dhcp conflict`
- `show ip dhcp server statistics`

C1

```
C1#
C1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.10.11       0198.fa9b.09c3.46       Oct 19 2025 05:30 PM    Automatic
192.168.20.11       0100.e04c.360f.cc       Sep 20 2025 05:27 PM    Automatic
C1#
C1#show ip dhcp server statistics
Memory usage         17715
Address pools        2
Database agents      0
Automatic bindings   2
Manual bindings      0
Expired bindings     0
Malformed messages   0
Secure arp entries   0
Renew messages       0

Message              Received
BOOTREQUEST          0
DHCPDISCOVER         3
DHCPREQUEST          2
DHCPDECLINE          0
DHCPRELEASE          0
DHCPINFORM           0

Message              Sent
BOOTREPLY            0
DHCPOFFER            2
DHCPACK              2
DHCPNAK              0
C1#
```

Hasil pengecekan pada L1 dan L2.

L1

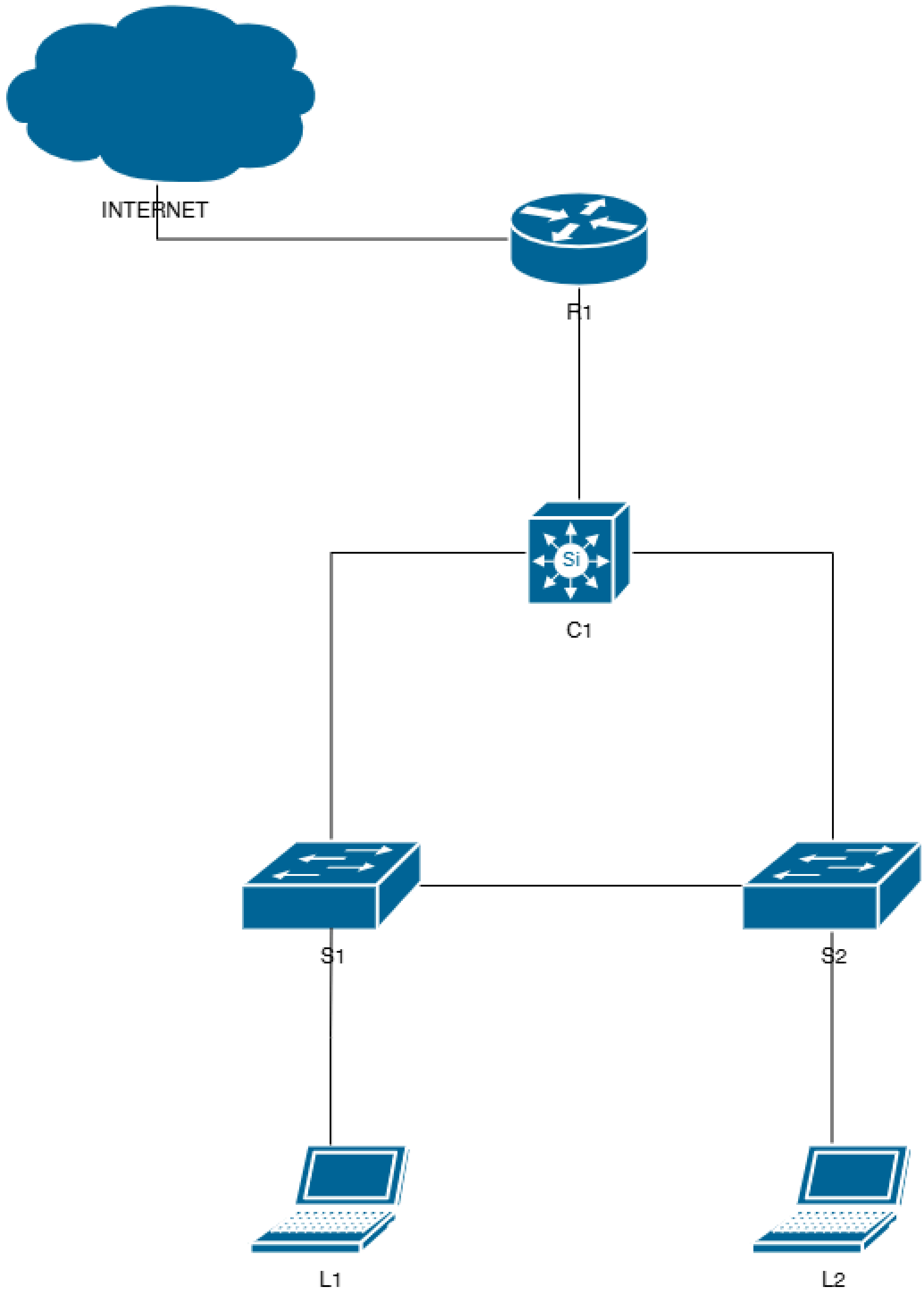


L2

User mendapatkan IP address DHCP sesuai dengan VLAN masing-masing.

Lab 6. RIP

Topology



Konfigurasi

- Konfigurasi RIP pada R1 dan C1

Untuk menghubungkan semua network, lakukan konfigurasi RIP pada switch L3 dan router.

R1

```
R1#
R1#show ip int brief
Interface                  IP-Address      OK? Method Status
Protocol
FastEthernet0/0            192.168.1.18    YES DHCP   up
up
FastEthernet0/1            10.1.2.1        YES manual up
up
Loopback0                  1.1.1.1         YES manual up
up
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 10.1.2.0
R1(config-router)#network 1.1.1.1
R1(config-router)#default-information originate
R1(config-router)#
```

C1

```
C1(config)#
C1(config)#router rip
C1(config-router)#version 2
C1(config-router)#no auto-summary
C1(config-router)#network 192.168.10.0
C1(config-router)#network 192.168.20.0
C1(config-router)#network 10.1.2.0
C1(config-router)#network 2.2.2.2
C1(config-router)#
```

Verifikasi

- Cek routing table di R1 dan C1
- Test ping end-to-end connection

R1

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

     1.0.0.0/32 is subnetted, 1 subnets
C       1.1.1.1 is directly connected, Loopback0
     2.0.0.0/32 is subnetted, 1 subnets
R       2.2.2.2 [120/1] via 10.1.2.2, 00:00:01, FastEthernet0/1
R    192.168.10.0/24 [120/1] via 10.1.2.2, 00:00:01, FastEthernet0/1
R    192.168.20.0/24 [120/1] via 10.1.2.2, 00:00:01, FastEthernet0/1
     10.0.0.0/24 is subnetted, 1 subnets
C       10.1.2.0 is directly connected, FastEthernet0/1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [254/0] via 192.168.1.1
R1#
```

C1

```
C1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.2.1 to network 0.0.0.0

     1.0.0.0/32 is subnetted, 1 subnets
R       1.1.1.1 [120/1] via 10.1.2.1, 00:00:14, FastEthernet1/0/1
     2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2 is directly connected, Loopback0
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
     10.0.0.0/24 is subnetted, 1 subnets
C       10.1.2.0 is directly connected, FastEthernet1/0/1
R*   0.0.0.0/0 [120/1] via 10.1.2.1, 00:00:14, FastEthernet1/0/1
C1#
```

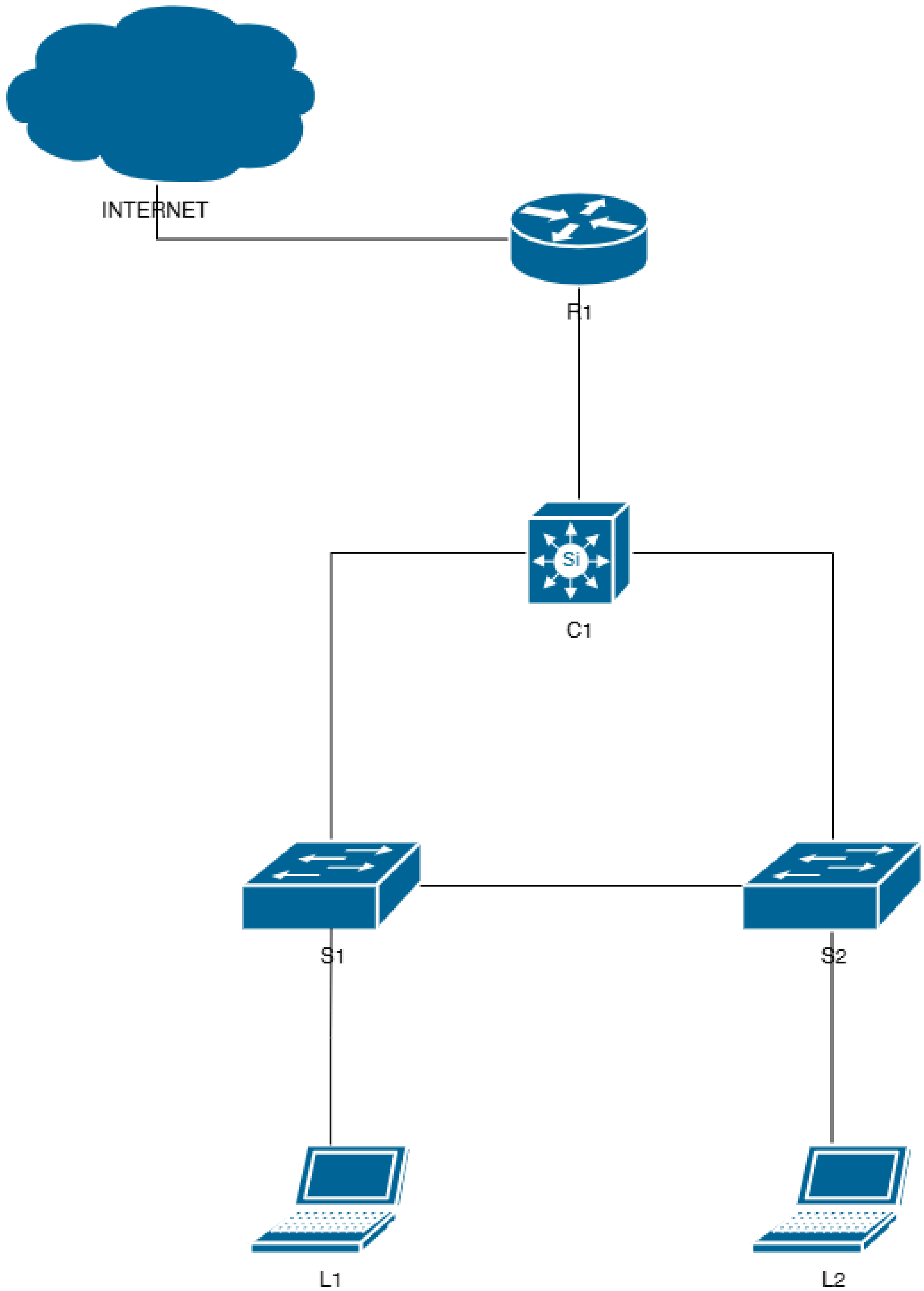
R1

```
R1#
R1#ping 192.168.10.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#
R1#ping 192.168.20.11

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R1#_
```

R1 berhasil ping ke L1 dan L2.

Lab 7. NAT

Topology

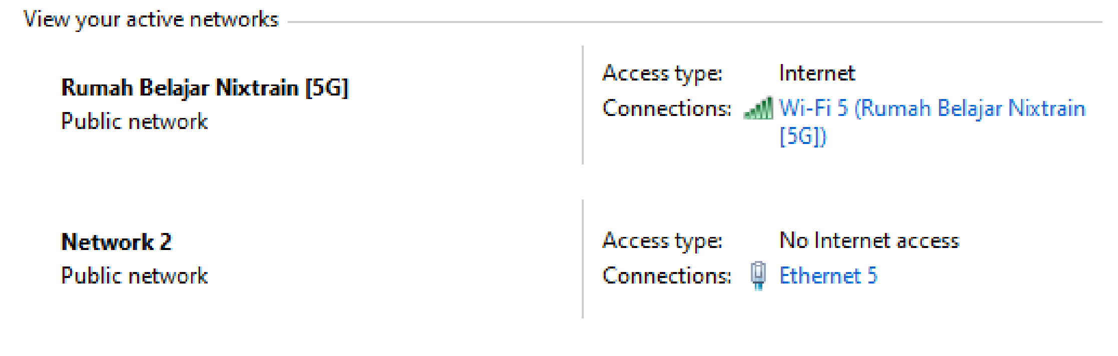


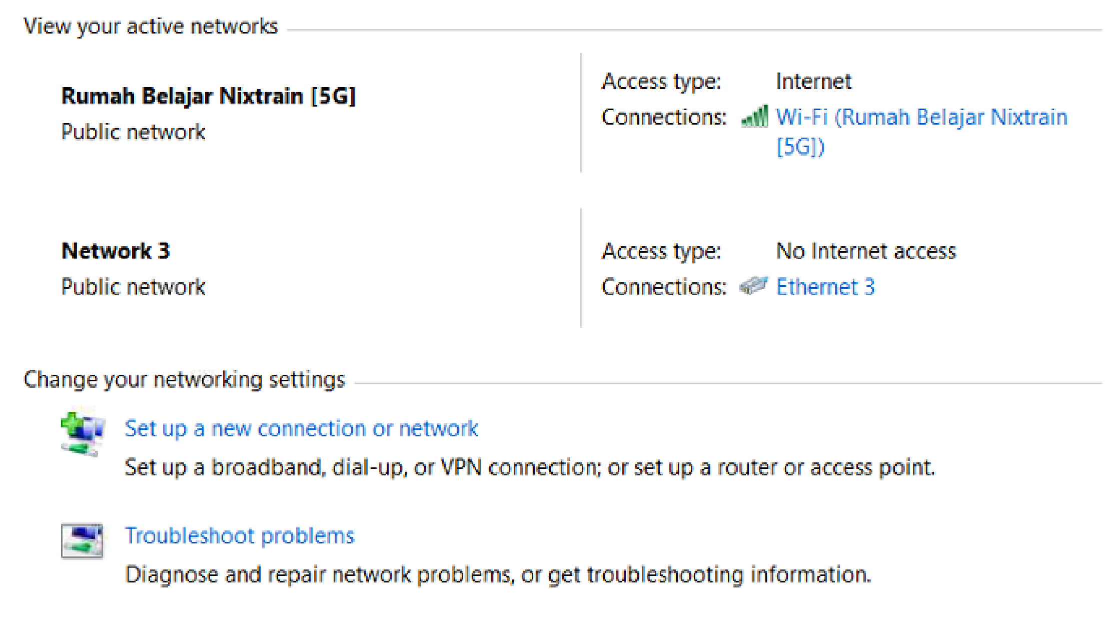
Konfigurasi

- Konfigurasi NAT pada R1

Agar semua network dapat terhubung ke internet, maka perlu dilakukan konfigurasi NAT di R1.

Status saat ini di L1 dan L2 : pada LAN access masih belum terhubung ke internet (***No Internet access***)

L1


L2


Lakukan konfigurasi NAT di R1.

R1

<pre>R1(config)#
R1(config)#access-list 1 permit 1.1.1.1 0.0.0.0
R1(config)#access-list 1 permit 2.2.2.2 0.0.0.0
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.20.0 0.0.0.255
R1(config)#access-list 1 permit 10.1.2.0 0.0.0.255
R1(config)#
R1(config)#int f0/1
R1(config-if)#ip nat inside
R1(config-if)#
R1(config-if)#int f0/0
R1(config-if)#ip nat outside
R1(config-if)#
R1(config-if)#ip nat inside source list 1 int f0/0 overload
R1(config)#</pre>

Verifikasi

Untuk melakukan verifikasi NAT, jalankan command berikut ini :

- `show ip nat translation`
- `show access-list`

R1

```
R1#
R1#show ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
icmp 192.168.1.18:1    192.168.20.11:1    216.239.38.120:1   216.239.38.120:1
udp 192.168.1.18:137   192.168.20.11:137  10.33.144.1:137    10.33.144.1:137
udp 192.168.1.18:137   192.168.20.11:137  125.160.0.105:137  125.160.0.105:137
udp 192.168.1.18:137   192.168.20.11:137  125.160.0.106:137  125.160.0.106:137
udp 192.168.1.18:137   192.168.20.11:137  192.168.1.1:137    192.168.1.1:137
udp 192.168.1.18:51478 192.168.20.11:51478 118.98.115.70:53  118.98.115.70:53
udp 192.168.1.18:51478 192.168.20.11:51478 118.98.115.77:53  118.98.115.77:53
udp 192.168.1.18:52308 192.168.20.11:52308 118.98.115.77:53  118.98.115.77:53
udp 192.168.1.18:52539 192.168.20.11:52539 118.98.115.70:53  118.98.115.70:53
udp 192.168.1.18:52539 192.168.20.11:52539 118.98.115.77:53  118.98.115.77:53
udp 192.168.1.18:54287 192.168.20.11:54287 118.98.115.70:53  118.98.115.70:53
udp 192.168.1.18:54287 192.168.20.11:54287 118.98.115.77:53  118.98.115.77:53
udp 192.168.1.18:56001 192.168.20.11:56001 118.98.115.70:53  118.98.115.70:53
udp 192.168.1.18:56001 192.168.20.11:56001 118.98.115.77:53  118.98.115.77:53
R1#
```

Traffic translation di router R1 sudah muncul. Berarti NAT sudah berjalan.

Dari sisi user L2, kita bisa lakukan traceroute ke google.com, pastikan path yang dipilih melalui R1.

L2

```
C:\Users\AGUS SETIAWAN>
C:\Users\AGUS SETIAWAN>tracert google.com

Tracing route to forcesafesearch.google.com [216.239.38.120]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  192.168.20.1
  2     1 ms     1 ms     1 ms  10.1.2.1
  3     2 ms     2 ms     2 ms  192.168.1.1
  4     3 ms     3 ms     3 ms  10.33.144.1
  5     *        4 ms     4 ms  125.160.0.106
  6     6 ms     5 ms     5 ms  125.160.0.105
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9    23 ms    23 ms    25 ms  180.240.205.82
 10    23 ms    22 ms    26 ms  72.14.213.104
 11    26 ms    34 ms    26 ms  192.178.109.207
 12    24 ms    26 ms    28 ms  142.251.241.3
 13    23 ms    25 ms    22 ms  any-in-2678.1e100.net [216.239.38.120]

Trace complete.

C:\Users\AGUS SETIAWAN>
```